

**ST ALOYSIUS (DEEMED TO BE UNIVERSITY), MANGALURU**  
**MID Term Examination – Semester I – September 2026**  
**BCA AI Augmented Software Systems**

**Subject: Algorithmic Thinking & Data Structures with C**

**Subject Code: 3D101**

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**Answer Key**

1. a)  $\frac{3}{16.00}$   
7

b)

```
#include <stdio.h>
```

```
int main() {
    int rollNumber;
    float marks1, marks2, marks3, marks4, marks5;
    float total, average, cgpa;

    // Input roll number and marks
    printf("Enter Roll Number: ");
    scanf("%d", &rollNumber);

    printf("Enter marks in 5 subjects:\n");
    printf("Subject 1: ");
    scanf("%f", &marks1);
    printf("Subject 2: ");
    scanf("%f", &marks2);
    printf("Subject 3: ");
    scanf("%f", &marks3);
    printf("Subject 4: ");
    scanf("%f", &marks4);
    printf("Subject 5: ");
    scanf("%f", &marks5);

    // Calculate total and average
    total = marks1 + marks2 + marks3 + marks4 + marks5;
    average = total / 5;

    // Convert average to CGPA
    cgpa = average / 9.5;

    // Display results
    printf("\n--- Student CGPA Report ---\n");
```

```
printf("Roll Number : %d\n", rollNumber);
printf("Total Marks : %.2f\n", total);
printf("Average    : %.2f\n", average);
printf("CGPA      : %.2f\n", cgpa);
```

```
return 0;
```

```
}
```

2. a)

Fizz

b)

```
#include <stdio.h>
```

```
int main() {
```

```
    float weight, distance;
```

```
    float baseCost, surcharge = 0, flatFee = 0, finalCharge;
```

```
    // Input package details
```

```
    printf("Enter weight of package (in kg): ");
```

```
    scanf("%f", &weight);
```

```
    printf("Enter distance to be shipped (in km): ");
```

```
    scanf("%f", &distance);
```

```
    // Calculate base cost
```

```
    baseCost = (weight * 10) + (distance * 2);
```

```
    // Apply 10% surcharge if weight exceeds 20 kg
```

```
    if (weight > 20) {
```

```
        surcharge = baseCost * 0.10;
```

```
    }
```

```
    // Apply flat fee of Rs.50 if distance exceeds 500 km
```

```
    if (distance > 500) {
```

```
        flatFee = 50;
```

```
    }
```

```
    // Calculate final shipping charge
```

```
    finalCharge = baseCost + surcharge + flatFee;
```

```
    // Display results
```

```
    printf("\n--- Shipping Charge Report ---\n");
```

```
    printf("Base Cost      : Rs.%.2f\n", baseCost);
```

```
    printf("Weight Surcharge : Rs.%.2f\n", surcharge);
```

```
    printf("Distance Flat Fee : Rs.%.2f\n", flatFee);
```

```
    printf("Final Shipping Charge : Rs.%.2f\n", finalCharge);
```

```

    return 0;
}

```

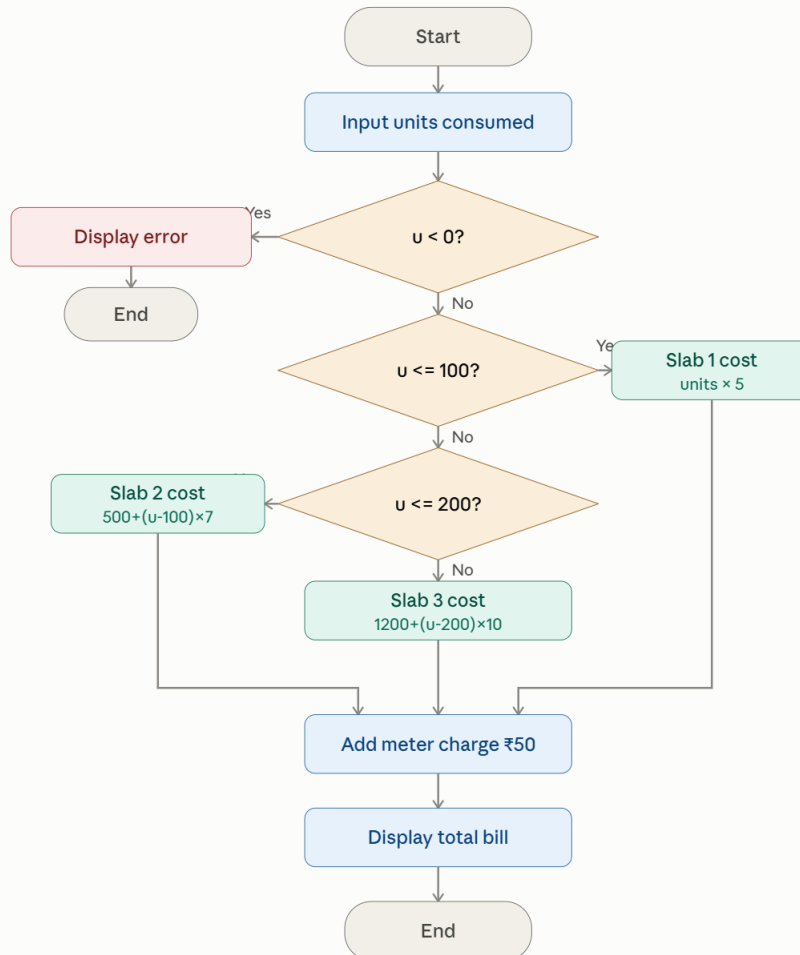
3. a)

```

Sum = 1
i value = 3

```

b)



START

DECLARE units, cost, meterCharge = 50, totalBill

INPUT units

IF units < 0 THEN

    PRINT "Error: Units cannot be negative"

ELSE

    IF units <= 100 THEN

        cost = units \* 5

    ELSE IF units <= 200 THEN

        cost = (100 \* 5) + (units - 100) \* 7

    ELSE

        cost = (100 \* 5) + (100 \* 7) + (units - 200) \* 10

    END IF

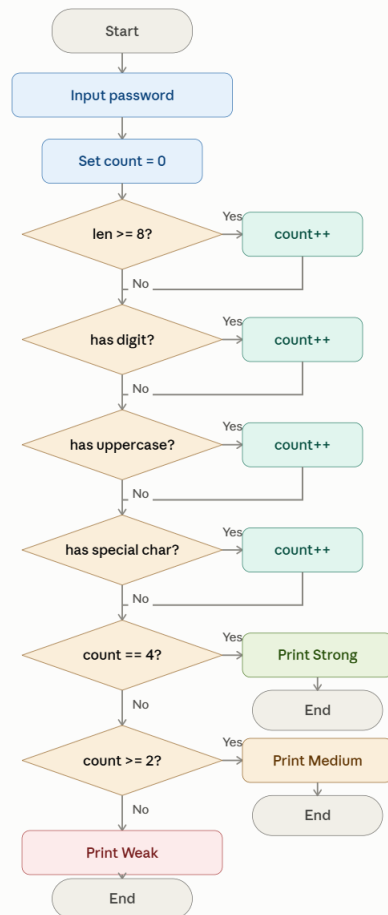
```

    totalBill = cost + meterCharge
    PRINT "Total Bill = ", totalBill
END IF

END

```

4. a)  
 Tax Category: B (Senior)  
 b)



START

DECLARE password (string), count = 0

INPUT password

```

IF length(password) >= 8 THEN
    count = count + 1
END IF

```

```

IF password contains at least one digit (0-9) THEN
    count = count + 1
END IF

```

```

IF password contains at least one uppercase letter (A-Z) THEN
    count = count + 1
END IF

```

```

IF password contains at least one special character (@, #, $, %, &, !) THEN
    count = count + 1
END IF

```

```

IF count == 4 THEN
    PRINT "Password Strength: Strong"
ELSE IF count >= 2 THEN
    PRINT "Password Strength: Medium"
ELSE
    PRINT "Password Strength: Weak"
END IF

```

```

END

```

5. a)

```

1 2 3 4 | 2 4 6 | 3 6 | 4 |
Count = 10

```

b)

```

#include <stdio.h>

```

```

int main() {
    int day, steps;
    int total = 0, goalAchievedCount = 0;
    int bestDaySteps = 0, bestDayNumber = 0;
    float average;
    const int GOAL = 8000;

```

```

    printf("Enter step count for 7 days:\n");

```

```

    for (day = 1; day <= 7; day++) {
        printf("Day %d: ", day);
        scanf("%d", &steps);

```

```

        // Accumulate total
        total += steps;

```

```

        // Check goal achievement
        if (steps >= GOAL) {
            goalAchievedCount++;
        }
    }

```

```
// Track the best day
if (steps > bestDaySteps) {
    bestDaySteps = steps;
    bestDayNumber = day;
}

average = total / 7.0;

// Display results
printf("\n--- Weekly Step Report ---\n");
printf("Total Steps      : %d\n", total);
printf("Average Steps    : %.2f\n", average);
printf("Goal Achieved Days: %d out of 7\n", goalAchievedCount);
printf("Best Day          : Day %d with %d steps\n", bestDayNumber,
bestDaySteps);

return 0;
}
```